

KABIR KAKKAR

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EDUCATION

Carnegie Mellon University

August 2024 - December 2025

Master of Information Systems Management

- **Coursework:** Distributed Systems, Database Management, Statistics, Agile, Machine Learning, Data Science for Product Managers, Artificial Intelligence (AI), Data Structures and Algorithms

Manipal University Jaipur

June 2017 - June 2021

Bachelor of Technology, Computer and Communication Engineering

SKILLS

Languages: Python, Java, SQL, C++, Javascript, Typescript

Backend & Systems: FastAPI, Flask, Django, Spring Boot, React, Node.js, Microservices, REST APIs, GraphQL

Data Engineering: Kafka, Spark (PySpark), Airflow, Hadoop, Spark Streaming

Databases & Warehousing: Snowflake, PostgreSQL, MongoDB, Redis, Pinecone (Vector DB)

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Terraform, CI/CD, Prometheus, Grafana, Jenkins

PROFESSIONAL EXPERIENCE

MedSarathi LLC

July 2025 - August 2025

Software Development Engineer, Internship

- Built an LLM-powered residency matching platform that extracted applicant data from uploaded documents and generated personalized program recommendations with 95%+ extraction accuracy.
- Developed a hybrid RAG recommendation engine across 100+ residency programs using semantic retrieval, custom ranking, and grounded LLM reasoning, delivering recommendations in under 3 seconds.
- Built scalable backend infrastructure for document ingestion, PostgreSQL storage, observability, and Dockerized deployment, supporting over 10,000 daily processing requests.

Accenture

May 2021 - May 2023

Software Engineer

- Developed backend services using asynchronous processing, caching, and database-query optimization, increasing request throughput by 35% and reducing p95 latency by 30% under peak load.
- Containerized backend services using Docker and deployed them on Kubernetes, improving deployment consistency across production environments.
- Built Jenkins-based CI/CD pipelines that automated build, test, Docker image creation, registry push, and Kubernetes deployment, reducing standard service deployment time from 4 hours to 12 minutes.
- Built reusable Terraform modules and Infrastructure-as-Code workflows, reducing environment provisioning time from hours to minutes while standardizing cloud deployments across 10+ services.
- Implemented Prometheus metrics, Grafana dashboards, and CloudWatch alerts across production environments, reducing mean time to detect production issues by 60%.

PROJECTS

Scalable Real-Time Recommendation Platform

- Designed and deployed a multi-stage recommendation system leveraging collaborative filtering and real-time re-ranking, improving CTR by 18% and NDCG@10 by 22% across personalized recommendations
- Built low-latency inference APIs handling 5K+ requests/sec with automated retraining pipelines and Grafana-based monitoring

Real-Time Anomaly Detection Pipeline for Financial Fraud

- Architected a production-grade anomaly detection pipeline using Apache Kafka and PySpark Structured Streaming to process high-velocity data streams, achieving a throughput of 1k+ events/sec
- Deployed an Isolation Forest model to identify fraudulent transactions with 92% recall and integrated concept drift monitoring to trigger automated alerts in Grafana when data distributions shifted

Research Paper: Implementation of Demand Forecasting – A Comparative Approach

- Co-authored a published comparative study evaluating demand forecasting models (Linear Regression vs. Gradient Boosted Trees) on real retail sales data, using RMSE and R^2 as evaluation metrics.
- Built an end-to-end forecasting pipeline with cloud deployment and Power BI dashboards to visualize predicted demand trends and support inventory decision-making.